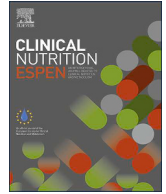




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Narrative Review

Nutrition during the menopausal transition: Current evidence, health trajectories and clinical implications

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SUMMARY

Background: The menopausal transition is a critical period in women's health, characterised by profound endocrine changes that may influence cardiometabolic risk, body composition, bone and muscle health, and overall functional trajectories. Beyond symptom burden, accumulating evidence suggests that this transition represents a window of vulnerability for the acceleration of long-term chronic disease risk, independently of chronological ageing.

Objective: This critical narrative review synthesises current evidence on the role of nutrition during the menopausal transition, focusing on cardiometabolic health, bone and muscle function, and the clinical interpretation of menopausal symptoms, with the aim of proposing a pragmatic and clinically translatable prevention framework.

Methods: A literature search was conducted in PubMed/MEDLINE, Embase and the Cochrane Library from inception to 2025. Priority was given to systematic reviews, meta-analyses, large longitudinal cohort studies and randomised controlled trials when available. Evidence was analysed qualitatively within a clinically oriented framework, with particular attention to heterogeneity of findings and limitations of nutritional interventions.

Results: The menopausal transition is associated, in some women, with unfavourable shifts in cardiometabolic trajectories, accelerated bone loss, and increased skeletal muscle vulnerability, although these changes remain heterogeneous and strongly influenced by lifestyle factors. The most consistent evidence suggests that overall dietary quality, particularly Mediterranean- and DASH-type dietary patterns, contributes to the modulation of cardiometabolic risk and supports bone and muscle health. In contrast, nutrition-specific effects on vasomotor symptoms appear modest and inconsistent. Nutritional strategies seem most relevant when implemented early, integrated with physical activity, and adapted to individual risk profiles.

Conclusion: Nutrition should be considered a modifiable component of lifestyle contributing to reasoned prevention during the menopausal transition. Its primary role lies in supporting functional preservation and modulating cardiometabolic trajectories rather than providing uniform symptomatic relief. An integrated, individualised approach embedded within broader medical and lifestyle strategies appears most clinically relevant at this life stage.

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1. Introduction

The menopausal transition is not limited to the cessation of reproductive function. It represents a period of profound endocrine changes, characterised by a progressive decline in oestrogen levels and complex hormonal fluctuations. These changes are accompanied by concomitant alterations in body composition,

lipid and glucose metabolism, vascular function, as well as bone and skeletal muscle health [1,2]. Evidence from large longitudinal cohort studies, notably the Study of Women's Health Across the Nation (SWAN), indicates that several cardiometabolic alterations tend to accelerate during this transition, independently of chronological ageing, suggesting the existence of a specific window of vulnerability in women's health trajectories [1].

In this review, the term "menopausal transition" refers to the perimenopausal period preceding the final menstrual period and

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the early years following its cessation, in accordance with commonly used clinical definitions.

The menopausal transition typically occurs between 45 and 55 years of age and may extend over several years before and shortly after the final menstrual period. It is characterized by increasing menstrual irregularity resulting from hormonal fluctuations and may be accompanied by vasomotor symptoms (hot flushes and night sweats), sleep disturbances, mood changes, fatigue, and changes in body composition. Although symptom presentation is highly heterogeneous, vasomotor symptoms are among the most commonly reported manifestations and may persist for several years beyond menopause [3].

From both a clinical and public health perspective, this period represents a major challenge. Systematic reviews and meta-analyses have shown that an earlier age at menopause is associated with an increased risk of cardiovascular disease, cardiovascular mortality and all-cause mortality [4]. These findings highlight the need to consider the menopausal transition not merely as an endocrine event, but as a critical period for the prevention of long-term chronic diseases.

In parallel, menopausal symptoms, particularly vasomotor symptoms, are still predominantly approached through their impact on quality of life. However, several syntheses of the literature suggest associations between the presence of vasomotor symptoms and a less favourable cardiometabolic profile, alterations in endothelial function, and an increased risk of mood disorders, although these associations remain heterogeneous and partly influenced by confounding factors [5]. Longitudinal data notably indicate a bidirectional relationship between vasomotor symptoms and depressive symptoms during the menopausal transition [6]. These observations invite a shift beyond a purely symptom-centred interpretation of menopause, towards considering certain symptoms as potential clinical markers of underlying vulnerability, while acknowledging the heterogeneity and limitations of the available evidence.

In this context, interest in nutritional interventions during menopause has increased substantially. Nevertheless, the nutrition literature remains fragmented, often structured around isolated nutrients, specific symptoms or individual risk factors, which may lead to reductionist or excessively prescriptive messages. This approach contrasts with clinical reality, in which women undergoing the menopausal transition frequently present with intertwined issues, including emerging cardiovascular risk factors, abdominal fat redistribution, declines in muscle mass and function, concerns regarding bone health, as well as sleep and mood disturbances.

The aim of this review is to examine the role of nutrition during the menopausal transition and its potential implications for cardiometabolic health, bone and muscle health, and menopausal symptoms, in order to provide a clinically oriented framework for prevention.

2. Methodological positioning and methods

This review is conducted as a critical narrative review, with the aim of providing a clinically oriented synthesis of the available evidence on the role of nutrition during the menopausal transition. This methodological approach was chosen in order to integrate data from heterogeneous fields, including endocrinology, cardiometabolic health, bone and muscle health, and clinical nutrition, which are difficult to combine within a single systematic review, while allowing for a contextualised interpretation that is directly translatable to clinical practice.

2.1. Literature search strategy

A literature search was conducted using the PubMed/MEDLINE, Embase and Cochrane Library databases, covering the period from their inception through 2025 (inclusive). The last literature search was performed in December 2025. This choice aimed to include the major historical data that have contributed to the understanding of health trajectories during the menopausal transition.

Search terms combined keywords related to menopause and the menopausal transition ("menopause", "menopausal transition", "perimenopause") with nutrition-related terms ("diet", "nutrition", "dietary patterns", "Mediterranean diet", "DASH diet") and outcome-related terms ("cardiovascular risk", "metabolic health", "bone health", "muscle mass", "sarcopenia", "vasomotor symptoms"). These terms were adapted to each database as appropriate. The reference lists of selected articles and relevant reviews were also screened to identify additional publications.

This review did not follow a PRISMA-based workflow and no formal study selection flow diagram was produced, as the objective was to provide a clinically integrative narrative synthesis rather than an exhaustive systematic mapping of the literature, which would have required a different methodological design.

2.2. Selection criteria and data prioritisation

Priority was given to

- systematic reviews and meta-analyses,
- large longitudinal cohort studies,
- randomised controlled trials when available and relevant.

Observational studies with a lower level of evidence, cross-sectional studies, and small-scale trials were included selectively when they provided relevant mechanistic, pathophysiological or clinical insights, particularly in areas where high-level evidence remains limited.

When multiple publications addressed the same topic, preference was given to studies with greater methodological robustness, adequate follow-up duration, larger sample sizes, and clinically meaningful outcomes (cardiovascular events, metabolic parameters, bone and muscle density or function, quality of life).

Study selection was guided by clinical relevance, methodological robustness, and consistency of findings across publications, rather than predefined quantitative eligibility thresholds.

The literature search, study selection, and qualitative analysis were conducted by the author, with the aim of clinically integrating the available evidence rather than pursuing exhaustive coverage.

When evidence specifically addressing women during the menopausal transition was limited, relevant data from broader populations (including postmenopausal women or mixed midlife cohorts) were considered and interpreted cautiously within a clinically oriented framework.

2.3. Critical appraisal and clinical interpretation

As this work is a narrative review, a formal grading of the level of evidence or strength of recommendations was not performed. No formal risk-of-bias assessment tool was applied systematically, in line with the objectives of this critical narrative review. Selected publications were analysed qualitatively, taking into account study design, consistency of findings across studies, clinical relevance of outcomes, and key methodological limitations, including residual

confounding, insufficient intervention duration, and heterogeneity of populations or interventions.

Data interpretation was deliberately focused on clinical translatability, distinguishing robust associations from hypotheses that remain debated, and explicitly acknowledging the limitations of nutritional interventions where applicable. The objective was not to provide an exhaustive inventory of the literature but rather to identify the most clinically informative and methodologically robust evidence relevant to health trajectories during the menopausal transition. Particular attention was paid to the timing of interventions, especially the distinction between the menopausal transition and postmenopause, in order to better address the concept of a critical window for prevention.

This approach was intended to ensure methodological coherence and transparency in interpretation, without claiming the exhaustiveness of a systematic review.

The narrative design was intentionally chosen to provide a clinically integrative perspective across heterogeneous fields and does not seek quantitative synthesis or reproducibility in the manner of a systematic review.

3. Menopausal transition: hormonal changes and health trajectories

The menopausal transition corresponds to a period of progressive endocrine remodeling, characterized by an early alteration of luteal function with a decline in progesterone secretion, related to an increased frequency of anovulatory cycles, followed by a fluctuating and subsequently sustained decrease in ovarian estradiol production. These hormonal changes are accompanied by a progressive rise in gonadotropins, particularly follicle-stimulating hormone (FSH), as well as a shift in the estrogen-androgen balance, with peripheral estrogen production becoming proportionally more prominent as ovarian function declines.

These changes extend far beyond reproductive function and exert systemic effects on multiple target tissues, including adipose tissue, skeletal muscle, bone, the cardiovascular system, and the brain. Data from large longitudinal cohort studies suggest that, in some women, this period is associated with an acceleration of certain cardiometabolic risk trajectories, independently of chronological aging [7].

3.1. Hormonal changes and metabolic desynchronization

Contrary to a simplified view of menopause as an abrupt and uniform decline in estrogen levels, data from longitudinal studies indicate that the menopausal transition is characterized by marked interindividual variability, with transient phases of relative hyper- and hypo-estrogenism [7,8]. Hormonal instability is accompanied by an early decline in progesterone secretion [3], related to the increasing frequency of anovulatory cycles, as well as by alterations in the estrogen-androgen balance [6], with peripheral estrogen production becoming proportionally more prominent as ovarian function declines.

From a metabolic perspective, this hormonal desynchronization is associated, in some women, with a preferential redistribution of fat mass toward the abdominal compartment, changes in lipid profile, and impaired insulin sensitivity [7]. The clinical expression of these changes remains highly variable and is strongly influenced by associated factors such as weight trajectory, body composition, physical activity level, pre-existing metabolic status, and lifestyle habits [9]. Data from large cohort studies suggest that these changes occur more frequently during the menopausal transition than during chronological aging alone,

supporting the hypothesis of a modulatory role of hormonal status in certain cardiometabolic risk trajectories, without a systematic or uniform pattern [1].

3.2. From hormonal changes to cardiometabolic risk trajectories

Longitudinal data suggest that the menopausal transition represents a period during which certain cardiometabolic risk trajectories may shift unfavorably, including blood pressure, lipid parameters, and glucose metabolism. These associations are not uniform, however, and occur within a broader context in which aging and lifestyle-related factors also play a major role [3,4].

Systematic reviews and meta-analyses of observational studies indicate that premature or early menopause is associated with an increased risk of cardiovascular events and mortality. In contrast, evidence regarding the independent role of time since menopause onset on cardiovascular outcomes remains limited and heterogeneous, precluding definitive conclusions about its long-term impact on cardiovascular risk, despite the biologically pivotal nature of the menopausal transition [2].

Taken together, these findings support the notion that menopause should not be considered an isolated risk factor, but rather as a potential accelerator of pre-existing cardiometabolic risk trajectories, shaped by genetic, behavioral, nutritional, and environmental determinants. From this perspective, prevention cannot be limited to late-stage management of established risk factors, but should instead be embedded within an anticipatory and individualized approach, ideally initiated during the menopausal transition.

3.3. Bone and muscle health and functional vulnerability

The menopausal transition is associated with an acceleration of bone loss, particularly pronounced during the first years following the decline in estrogen production. Longitudinal studies show that this phase corresponds to a period of rapid loss of bone mineral density, with medium-term consequences for fracture risk [4,10].

In parallel, declining estrogen levels may contribute to increased vulnerability of skeletal muscle tissue, without systematically leading to overt impairments in muscle mass, strength, or physical performance around the menopausal transition [11–13]. Available evidence suggests that the clinical expression of this vulnerability is largely influenced by factors such as age, physical activity level, protein intake, and metabolic context [14,15]. In this regard, the menopausal transition may represent a period of heightened sensitivity to the deleterious effects of sedentary behavior, insufficient nutritional intake, or comorbid conditions, with potential long-term implications for physical function and autonomy.

These observations underscore the relevance of considering bone and muscle health as a functional continuum, and of prioritizing the maintenance of function as a central clinical objective from the menopausal transition onward, rather than waiting for the emergence of late manifestations such as frailty or established sarcopenia.

3.4. Symptoms and health trajectories: beyond immediate symptom burden

Vasomotor symptoms, often regarded as the most visible manifestation of the menopausal transition, reflect a complex neuroendocrine dysregulation of central thermoregulation, involving hypothalamic pathways and their interactions with the autonomic nervous system. These symptoms may also be modulated by contextual factors such as stress and neurovegetative

activation [16]. Associations between the presence of vasomotor symptoms and cardiometabolic, vascular, or psychological health parameters have been reported; however, their heterogeneity and predominantly observational nature preclude establishing causal relationships or defining their precise clinical significance at this stage.

These elements invite clinicians and researchers to consider symptoms not only as targets for symptomatic relief, but also as potential clinical signals embedded within broader health trajectories, thereby justifying a comprehensive and individualized assessment of women during the menopausal transition.

4. Cardiometabolic health: nutrition as a modulator of risk trajectories

The menopausal transition may be accompanied, in some women, by an unfavourable shift in the cardiometabolic profile, characterised in particular by a deterioration in lipid metabolism during the transition to postmenopause, while other parameters (adipose tissue distribution, glycaemic regulation, blood pressure) tend to evolve more progressively and through multifactorial mechanisms [8]. These changes are neither uniform nor inevitable but rather occur within risk trajectories shaped by genetic, behavioural, and environmental determinants. In this context, nutrition should not be considered as an isolated therapeutic intervention, but rather as a lever for risk modulation, integrated within a comprehensive preventive strategy.

4.1. Dietary patterns and cardiovascular risk during menopause

Mediterranean-type dietary patterns are characterized by a high intake of fruits, vegetables, legumes, whole grains, olive oil, and fish, together with a reduced consumption of ultra-processed foods and saturated fats [17], whereas the DASH model emphasizes a plant-rich and low-sodium diet, primarily aimed at reducing blood pressure [18].

The most robust available evidence relates to overall dietary patterns rather than isolated nutrients. Several systematic reviews and meta-analyses show that higher adherence to Mediterranean or DASH-type dietary patterns is associated with a significant reduction in cardiometabolic risk, including type 2 diabetes, as well as lower cardiovascular and all-cause mortality, particularly among middle-aged and postmenopausal women [19,20]. These benefits are mainly driven by improvements in lipid profile, reductions in blood pressure, and better glycaemic regulation, without being attributable to any single dietary component [19].

Among women undergoing the menopausal transition, cohort studies highlight a specific period of cardiometabolic vulnerability, notably characterised by a transient deterioration in lipid profile during the late menopausal transition [1].

4.2. Nutrition, glucose metabolism and insulin resistance: a contextualised approach

As discussed previously, impairment of insulin sensitivity is neither systematic nor exclusively attributable to menopause. Nevertheless, among women presenting with abdominal fat redistribution, overweight, or a family history of type 2 diabetes, the menopausal transition may represent a period of increased metabolic vulnerability [8,21,22].

Systematic reviews indicate that nutritional interventions focusing on improving the quality of carbohydrate intake—through reductions in refined carbohydrates and increased dietary fibre from minimally processed or unprocessed foods—as well as

on overall dietary balance, are associated with modest but statistically significant improvements in glycaemic parameters [23,24].

However, these effects appear to be strongly dependent on baseline metabolic status, physical activity level, and long-term adherence, highlighting the importance of an individualised and contextualised approach.

4.3. Blood lipids: expected effects and limitations of nutrition

Changes in the lipid profile observed during the menopausal transition, particularly increases in LDL cholesterol, do not result from a single or uniform mechanism. Rather, they reflect heterogeneous metabolic trajectories that are at least partly independent of chronological ageing [1].

In this context, meta-analyses of randomized controlled trials indicate that nutritional interventions can contribute to modest but clinically meaningful improvements in lipid profiles, particularly through a reduction in saturated fatty acids in favor of cis-unsaturated fatty acids, which is associated with a decrease in LDL cholesterol and in the total cholesterol/HDL cholesterol ratio [25].

In addition, other meta-analyses indicate that a diet higher in dietary fiber and minimally processed plant-based foods, including in particular legumes and certain whole grains, is associated with a modest but statistically significant reduction in LDL cholesterol [26].

However, these effects remain variable and, on their own, insufficient to normalize a markedly altered lipid profile. This limitation should be clearly stated in order to avoid any over-interpretation of the role of nutrition, which here falls within a framework of primary prevention or risk reduction, rather than substitution for pharmacological treatments when these are indicated.

4.4. Body weight, body composition, and cardiometabolic risk

Weight gain frequently attributed to menopause largely reflects age-related factors, lifestyle changes, and a decline in physical activity. Nevertheless, in some women, the menopausal transition is accompanied by a preferential redistribution of fat mass toward the abdominal compartment, independent of total body weight, thereby contributing to an increased cardiometabolic risk [8].

Available data suggest that a lifestyle-based approach primarily targeting weight stabilization and reduction of abdominal adiposity (notably assessed by waist circumference) may improve cardiometabolic risk factors, even in the absence of substantial weight loss [27].

This pragmatic strategy, centered on measurable objectives, appears particularly relevant during the menopausal transition, a period characterized by heterogeneous cardiometabolic trajectories [1].

4.5. Role of nutrition within an integrated prevention strategy

Overall, nutrition appears to act as a modulator of cardiometabolic trajectories, with its impact closely dependent on the timing of intervention, the clinical profile, and the life context of each woman. Its potential seems greatest when implemented early, ideally during the early stages of the menopausal transition before the establishment of adverse cardiometabolic trajectories, and when combined with regular physical activity, including both aerobic and muscle-strengthening activities, as well as the management of other risk factors.

It therefore appears essential to reposition nutritional interventions not as uniform responses to hormonal changes, but as

tools for personalized prevention, aimed at limiting the acceleration of unfavorable cardiometabolic trajectories in women at higher risk.

5. Bone and muscle health: nutrition in support of functional maintenance

The menopausal transition represents a critical period for bone health, marked by a transient acceleration of bone remodeling related to the decline in estrogen production. Longitudinal studies show that this phase concentrates most of the bone loss, occurring around the time of the final menstrual period, and contributes to an increased subsequent risk of osteoporosis and fractures [10,26]. However, individual bone trajectories remain heterogeneous and are strongly influenced by modifiable factors, among which nutrition and physical activity play a central role.

5.1. Bone health: nutritional intakes, prevention, and limitations

Systematic reviews and meta-analyses indicate that adequate calcium and vitamin D intakes are necessary for the maintenance of bone health in menopausal women, particularly in the presence of deficiency or insufficiency. However, available evidence shows that calcium and vitamin D supplementation alone is not sufficient to prevent fractures in all women, underscoring the need for an integrated approach [28,29].

Beyond these micronutrients, overall diet quality appears to be a key determinant.

Mediterranean-type dietary patterns have been associated with higher bone mineral density and a reduced risk of fractures in several observational studies and narrative syntheses, although interventional data remain limited. These effects appear to be mediated by higher intakes of fruits and vegetables, high-quality protein sources, and unsaturated fatty acids, as well as by a reduction in overall inflammatory burden [30].

5.2. Muscle, sarcopenia, and functional vulnerability

Contrary to a common assumption, the menopausal transition does not systematically lead to a loss of muscle mass or muscle strength. However, available evidence suggests that the decline in oestrogen levels may contribute to increased vulnerability of skeletal muscle through effects on protein synthesis, muscle tissue composition, and adaptation to exercise. Emerging evidence also suggests that declines in muscle strength may occur earlier than measurable reductions in muscle mass during the menopausal transition, highlighting the importance of functional outcomes when assessing early muscle vulnerability. The clinical expression of this vulnerability largely depends on factors such as age, level of physical activity, protein intake, and the underlying metabolic context. Available expert consensus statements suggest that protein intakes of approximately 1.0–1.2 g/kg/day, combined with regular resistance exercise, may help support muscle mass and function in ageing adults [14]. However, specific evidence focused exclusively on women during the menopausal transition remains limited. In addition, oestrogen decline may contribute to reduced anabolic sensitivity to dietary protein, reinforcing the relevance of combining adequate protein intake with resistance exercise during this period [31].

Adequate protein intakes, distributed throughout the day, are associated with the maintenance of muscle mass and function, particularly when combined with regular physical activity including resistance training. From a practical perspective, these findings support the importance of ensuring sufficient protein intake at each main meal and of combining nutritional strategies

with regular resistance exercise in order to optimise muscle protein synthesis during the menopausal transition. In menopausal women, such strategies appear more relevant than isolated nutritional interventions aimed at increasing muscle mass, as the primary objective is the preservation of function and physical performance.

5.3. From mass to function: a primary clinical objective

Clinical relevance does not lie solely in the preservation of bone or muscle mass, but rather in the maintenance of function, including strength, balance, mobility, and the ability to perform activities of daily living. Longitudinal studies indicate that functional performance is a stronger predictor of morbidity and loss of autonomy than body mass alone [32].

In this context, nutrition should be considered as a support for the maintenance of function, in synergy with physical activity, rather than as an isolated corrective strategy. This approach allows nutritional objectives to be reframed around clinically meaningful outcomes, particularly during the menopausal transition, a period in which early interventions may help limit the development of longer-term frailty trajectories.

5.4. Limitations of nutritional interventions and integration with other strategies

It is essential to emphasize that nutritional interventions, although necessary, are not sufficient on their own to prevent osteoporosis, sarcopenia, or functional decline. Their effectiveness depends on the clinical context, long-term adherence, and their integration with other strategies, particularly adapted physical activity and, when indicated, pharmacological treatments.

Acknowledging these limitations contributes to a more realistic approach to care and helps avoid oversimplified or excessively prescriptive narratives. It also allows nutrition to be repositioned within a comprehensive strategy for prevention and functional maintenance, consistent with the available scientific evidence.

6. Vasomotor symptoms: role of nutrition and clinical interpretation

Vasomotor symptoms are among the most frequent and visible manifestations of the menopausal transition. They are traditionally addressed primarily in terms of their impact on quality of life, particularly because of their effects on sleep, daytime fatigue, and psychological well-being. While this perspective is legitimate, it tends to isolate symptoms from their broader pathophysiological and clinical context [5].

6.1. Vasomotor symptoms and health trajectories

Several studies suggest that the presence and severity of vasomotor symptoms may be embedded within broader health trajectories, particularly cardiometabolic and psychological ones. Associations have been reported between vasomotor symptoms, endothelial dysfunction, less favorable cardiometabolic profiles, and depressive symptoms. The relationship between vasomotor symptoms and mental health appears to be bidirectional, highlighting the complexity of the underlying mechanisms [33].

However, it should be emphasized that these associations do not allow the establishment of a direct causal relationship. Vasomotor symptoms should therefore be considered as potential clinical signals, possibly indicating underlying vulnerability, rather than as independent risk factors or isolated therapeutic targets.

6.2. Role of nutrition in the management of vasomotor symptoms

Interest in nutritional interventions aimed at reducing vasomotor symptoms has increased substantially, particularly with regard to specific foods, nutrients, or plant-derived supplements. From a clinical perspective, it is useful to distinguish between evidence related to overall dietary patterns, individual foods, and specific bioactive compounds, particularly phytoestrogens, which represent the main focus of most intervention studies in this area. However, evidence from systematic reviews and meta-analyses indicates that the effects of these nutritional approaches on the frequency and severity of vasomotor symptoms are overall modest, heterogeneous, and often of limited clinical relevance [34,35].

Nutritional approaches may contribute, in some women, to a partial improvement in symptom-related comfort when implemented within a high-quality overall dietary pattern and an integrated lifestyle approach that includes physical activity and broader lifestyle measures. However, nutrition cannot be considered either an alternative to hormone therapy when indicated or a universal solution for vasomotor symptoms [36].

6.3. Symptoms, patients' expectations, and the risk of overinterpretation

The strong expectations of patients regarding non-hormonal solutions expose both patients and clinicians to a risk of overinterpreting the benefits of nutritional interventions and to the dissemination of oversimplified messages that may be insufficiently supported by the available level of evidence. This situation may lead to restrictive strategies that are difficult to sustain over the long term, or to excessive self-blame when symptoms persist.

In this context, the role of healthcare professionals is to reposition nutrition within a realistic framework, clearly distinguishing what relates to global health support from what pertains to symptomatic treatment. A nuanced approach makes it possible to address patients' expectations while avoiding the promotion of unfounded promises.

6.4. Toward an integrated interpretation of symptoms

Considering vasomotor symptoms from an integrated perspective makes it possible to move beyond an exclusively symptom-focused approach and to embed their management within a broader prevention strategy. This perspective encourages a comprehensive assessment of associated risk factors, including lifestyle, diet, sleep, stress, and mental health, and opens the way to personalized interventions tailored to each woman's clinical profile and individual priorities [37].

The conceptual framework summarizing the main health trajectories associated with the menopausal transition and the potential role of nutrition and integrated lifestyle strategies is presented in Fig. 1.

Nutrition is presented as a modifiable lifestyle component that may contribute to the modulation of cardiometabolic trajectories, bone health, muscle function and, although more modestly and inconsistently, vasomotor symptoms, within an integrated strategy that also includes physical activity, sleep and stress management, and medical treatment when indicated.

7. Defining priorities: toward a pragmatic framework for nutritional management during the menopausal transition

The preceding sections highlight the complexity of interactions between hormonal changes, health trajectories, and lifestyle

determinants during the menopausal transition. In this context, an effective nutritional approach cannot be uniform, nor can it focus on a single symptom or isolated risk factor. Instead, it requires a clear prioritization of objectives, grounded in the level of available evidence and in medium- and long-term clinical relevance.

7.1. Prioritizing nutritional objectives

During the menopausal transition, nutritional priorities can be organized into three complementary levels.

The first priority concerns the preservation of function, including muscle and bone function and, more broadly, overall functional capacity. Maintaining mobility, autonomy, and physical function represents a central objective, given its major impact on quality of life and on the long-term risk of frailty. From this perspective, nutrition primarily aims to ensure sufficient and balanced intakes, in coherence with physical activity, rather than to correct isolated biological markers [28–32].

The second priority relates to the modulation of cardiometabolic trajectories. In women presenting with early risk factors or increased vulnerability, dietary interventions represent a relevant lever to limit the progression of cardiovascular and metabolic risk. This objective fits within a framework of primary prevention or risk reduction and does not replace pharmacological treatments when these are indicated [19,26].

The third priority concerns symptom management, particularly vasomotor symptoms. Although nutrition may contribute, in some women, to partial improvement in symptom-related discomfort, it should not be presented as a universal solution or as a systematic alternative to validated treatments. This hierarchical approach helps avoid excessive focus on symptoms at the expense of longer-term health priorities [33,37].

This framework may help clinicians distinguish whether cardiometabolic risk modulation, functional preservation, or symptom-related support should be considered the primary objective of nutritional intervention during the menopausal transition.

7.2. Identifying clinical profiles for which nutrition is particularly relevant

Building on this hierarchical framework of objectives, it appears that the relevance and impact of nutritional interventions vary according to clinical profile and timing of implementation. Nutrition therefore seems particularly relevant in women presenting with abdominal fat redistribution, early cardiometabolic risk factors, reduced physical activity, or insufficient or unbalanced dietary intakes. In these situations, it represents an accessible lever to modify certain risk trajectories before the development of established pathologies [1,9,21,22].

Conversely, when nutritional intervention is implemented at a later stage, in the context of already established diseases or in the absence of adherence to lifestyle changes, its impact appears more limited and should not be expected to function as an isolated therapeutic lever.

This differentiated approach allows nutrition to be embedded within a targeted and realistic strategy, to tailor messages to individual clinical priorities, and to avoid generic recommendations with limited effectiveness. It thereby strengthens the clinical relevance and translatability of nutritional interventions during the menopausal transition.

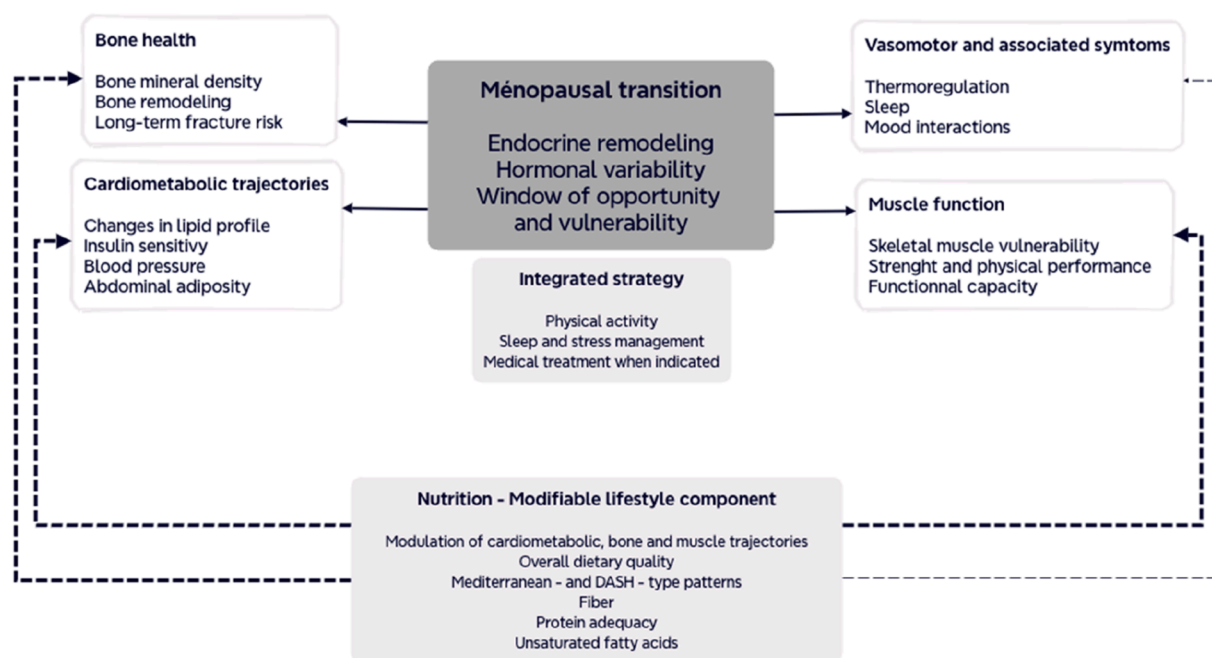


Fig. 1. Conceptual framework illustrating health trajectories during the menopausal transition and the potential role of nutrition.

7.3. Clarifying the role of nutrition within an integrated prevention strategy

From this perspective, nutrition cannot be considered as an isolated intervention. Its effectiveness is closely dependent on its integration into a comprehensive strategy that includes physical activity, sleep and stress management, and, when indicated, hormonal or pharmacological treatments. The menopausal transition represents, in this regard, a particularly favourable window of opportunity to initiate or reinforce

these interventions in a coordinated manner, before the establishment of risk trajectories that are more difficult to modify [1,9].

Clarifying the true role of nutrition also helps to prevent the drift associated with oversimplified or guilt-inducing narratives, often centred on excessive promises. This clarification contributes to repositioning nutritional care as a tool for reasoned prevention, grounded in the available scientific evidence, integrated within a multimodal approach, and adapted to the clinical priorities and individual expectations of women [37].

Table 1

Clinical objectives, expected benefits, limitations, and proposed mechanisms of nutritional interventions during the menopausal transition.

Clinical domain	Clinical objective/targeted trajectory	Nutritional approach	Potential benefits	Limitations/strength of clinical inference	Proposed mechanisms
Cardiometaabolic health	Limit unfavorable inflection of risk trajectories	Mediterranean/DASH dietary patterns	Improvement in lipid profile, blood pressure, and glycemic control	Modest effects, dependent on clinical context and adherence	Overall diet quality, dietary fiber, unsaturated fatty acids, reduction of inflammation
Glucose metabolism	Reduce insulin vulnerability	Carbohydrate quality, dietary fiber, minimally processed foods	Modest improvement in glycemic parameters	Variable effects depending on metabolic profile and physical activity	Improved insulin sensitivity, reduced glycemic load
Blood lipids	Modulate lipid trajectory	Reduction of saturated fatty acids, increased cis-unsaturated fatty acids	Reduction in LDL-cholesterol	Insufficient effects alone in cases of significant dyslipidemia	Modulation of hepatic lipid metabolism
Bone health	Preserve bone capital	Adequate calcium and vitamin D intake, overall high-quality diet	Support of bone mineral density	Isolated supplementation often insufficient to prevent fractures	Bone remodeling, calcium homeostasis
Muscle health	Maintain function and autonomy	Adequate protein intake + physical activity	Maintenance of muscle mass and function	Requires concomitant resistance exercise	Stimulation of muscle protein synthesis
Vasomotor symptoms	Improve symptom comfort without overpromising	Overall dietary approach (possibly including specific plant-based components)	Partial improvement possible in some women	Heterogeneous data, generally modest effects	Neuroendocrine modulation and metabolic factors

7.4. What nutrition can and cannot do during the menopausal transition

What nutrition can do

- Contribute to the modulation of cardiometabolic risk trajectories, particularly when implemented early in women with emerging risk factors.
- Support the maintenance of muscle and bone function, in synergy with physical activity, notably through adequate protein intake and an overall high-quality diet.
- Contribute, in some women, to a partial improvement in symptomatic comfort, particularly vasomotor symptoms, within the context of an integrated lifestyle approach.
- Act as a lever for reasoned prevention, promoting sustainable behaviours compatible with long-term health.

What nutrition cannot do;

- Reverse the hormonal changes inherent to the menopausal transition.
- Replace hormonal or pharmacological treatments when these are indicated.
- Guarantee a systematic or uniform reduction of symptoms, particularly vasomotor symptoms.
- Correct established pathologies on its own or compensate for a lack of adherence to lifestyle changes.
- Justify restrictive, guilt-inducing, or excessively prescriptive approaches that are disconnected from the available level of evidence.

The main clinical objectives, expected benefits, limitations, and proposed mechanisms of nutritional interventions during the menopausal transition are summarized in [Table 1](#).

Data synthesised from the evidence discussed in references [5,10,19–37], including systematic reviews, meta-analyses, cohort studies and expert consensus statements cited throughout the manuscript.

8. Conclusion and clinical perspectives

This review highlights that the menopausal transition represents a pivotal period in women's health trajectories, during which nutrition may act as a modifiable lifestyle factor influencing cardiometabolic risk, bone and muscle health, and overall well-being.

Available evidence indicates that these changes are neither uniform nor inevitable but rather unfold along heterogeneous trajectories shaped by individual, behavioral, and environmental determinants.

In this context, nutrition should not be considered as an isolated intervention or as an exclusively symptom-oriented response, but rather as a modifiable lifestyle component that may contribute to reasoned prevention. Its impact appears to depend closely on the timing of intervention, baseline risk profile, and its integration within a comprehensive management strategy. The most consistent available evidence suggests that overall dietary quality, particularly Mediterranean and DASH dietary patterns, contributes to the modulation of cardiometabolic risk and supports bone and muscle health, while effects on menopausal symptoms remain overall modest and heterogeneous.

Nutritional interventions cannot replace other pillars of prevention, including regular physical activity, sleep and stress management, or hormonal and pharmacological treatments when indicated. Recognizing both the potential and the limits of

nutrition is essential to avoid reductionist narratives and unrealistic expectations.

The menopausal transition may therefore be considered a window of opportunity to initiate or reinforce lifestyle strategies that are likely to influence long-term health trajectories. However, further well-designed longitudinal and interventional studies are needed to better identify the women most likely to benefit from targeted nutritional approaches and to clarify optimal implementation strategies.

Pending these data, an individualized, evidence-informed, and multidisciplinary approach remains the most clinically relevant framework to support women during the menopausal transition.

Declaration of generative AI and AI-assisted technologies

During the preparation of this manuscript, the author used ChatGPT5 (OpenAI) to assist with language editing and translation of the text. Following the use of this tool, the author carefully reviewed, revised, and edited the content as needed and takes full responsibility for the integrity, accuracy, and originality of the manuscript.

Declaration of interests

Marie-Caroline Baraut reports having had professional activities (consulting and educational missions) for Danone and Lactel in the past. These activities were not related to the subject of the present manuscript. The author declares that this work was conducted independently, without influence from any commercial entity.

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